

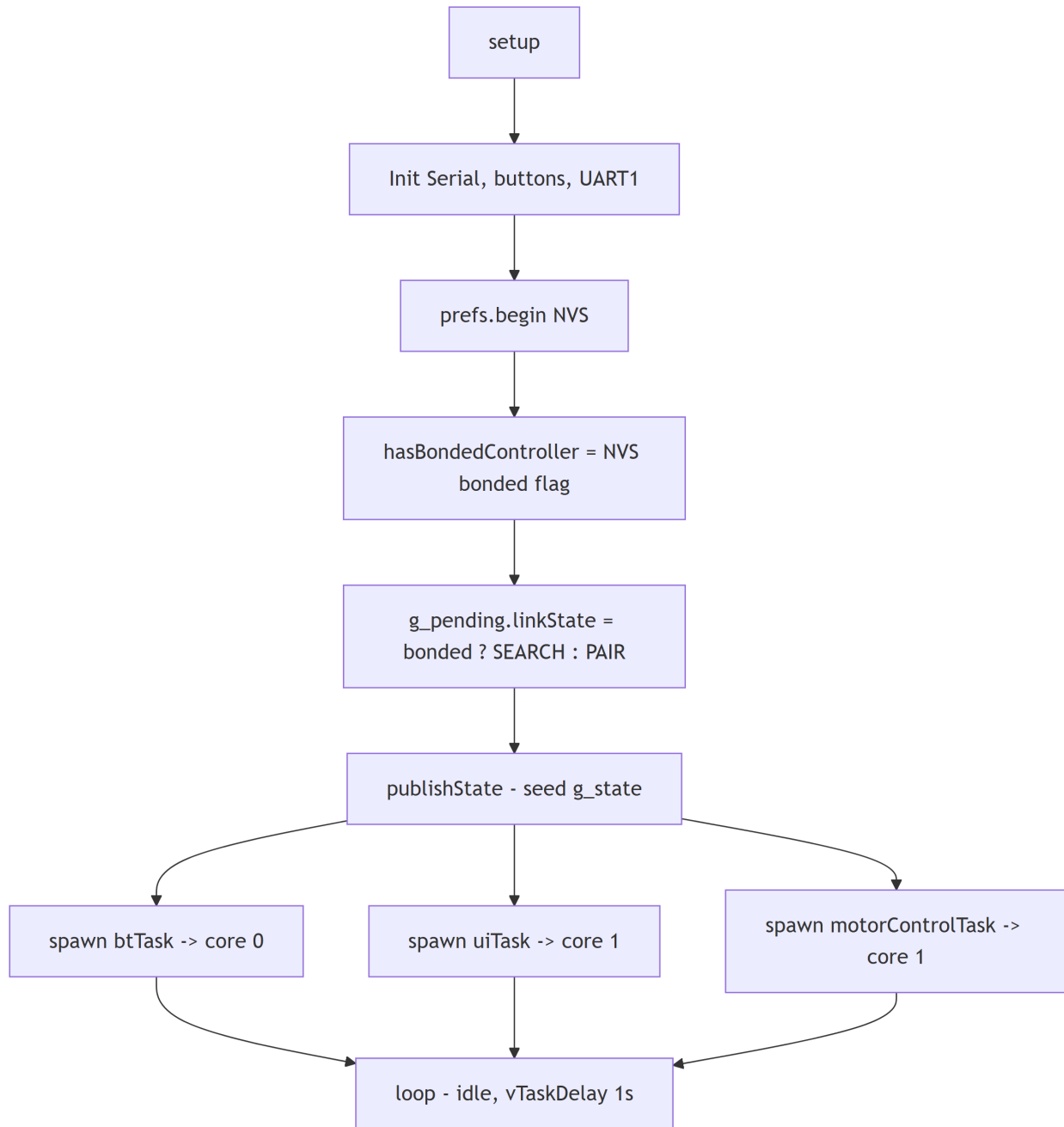
NitroWorks ECU Firmware

Control Flow Reference

Boot sequence, per-task loops, link-state machine, and the gamepad-to-motor data path.
Generated from docs/03_control_flow.md — sprint-1

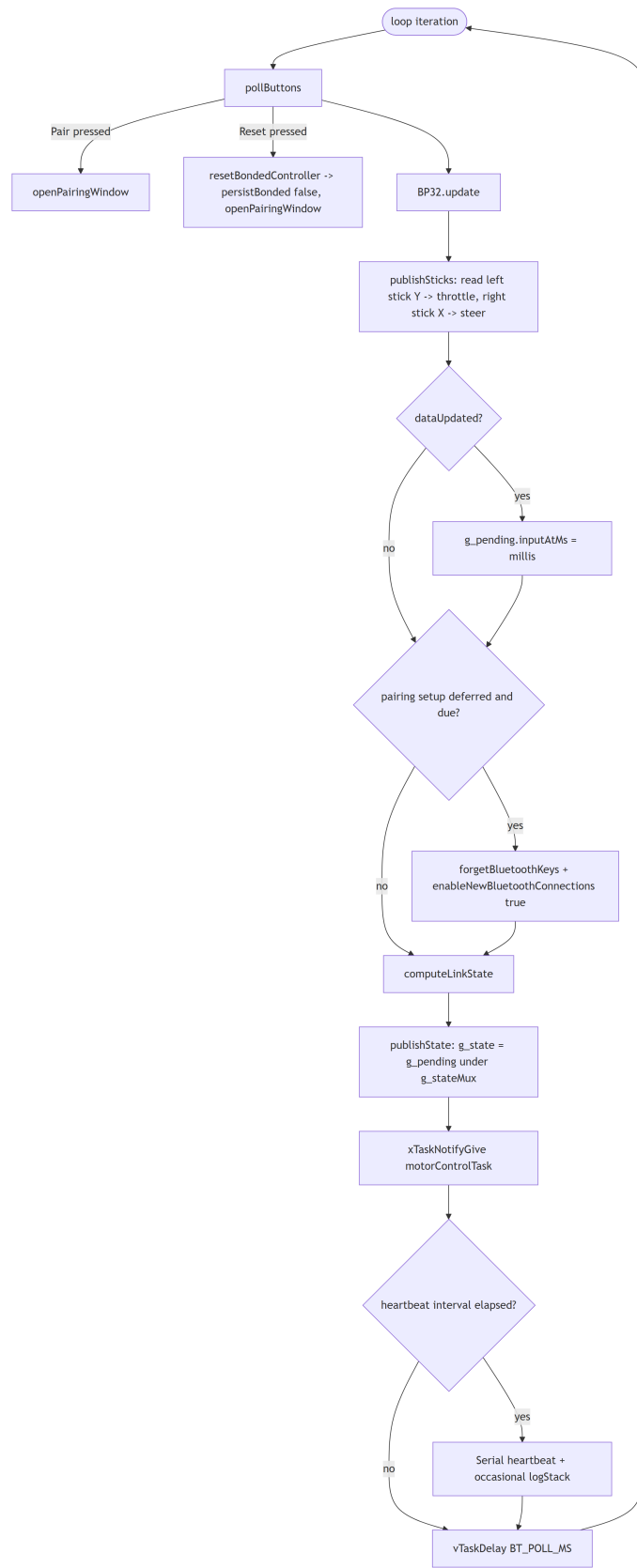
Boot Sequence

setup() in main.cpp loads the NVS bonded flag, seeds the shared RobotState via publishState(), then spawns all three pinned FreeRTOS tasks before falling into an idle loop().



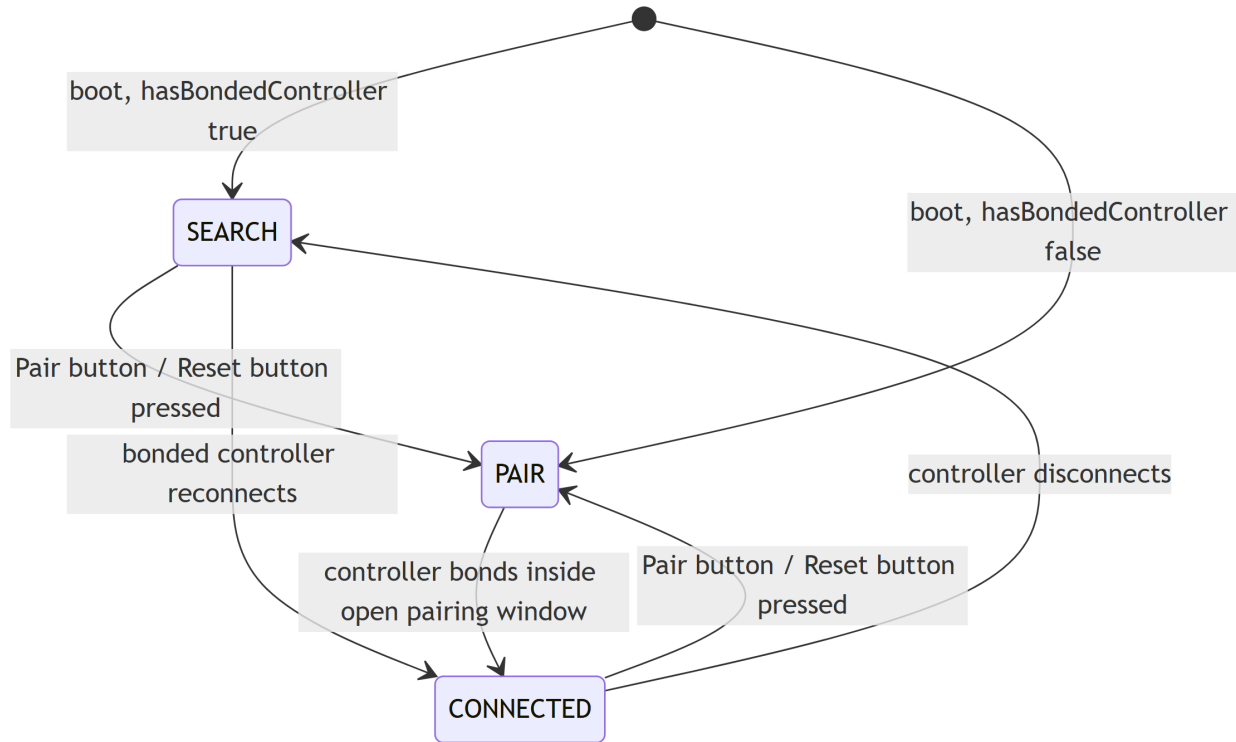
btTask Loop (core 0, every BT_POLL_MS = 5 ms)

Polls the Pair/Reset buttons, updates Bluepad32, reads the gamepad's left-stick Y (throttle) and right-stick X (steer) into g_pending, runs any deferred pairing key-wipe, recomputes link state, publishes the frame under g_stateMux, and wakes motorControlTask.



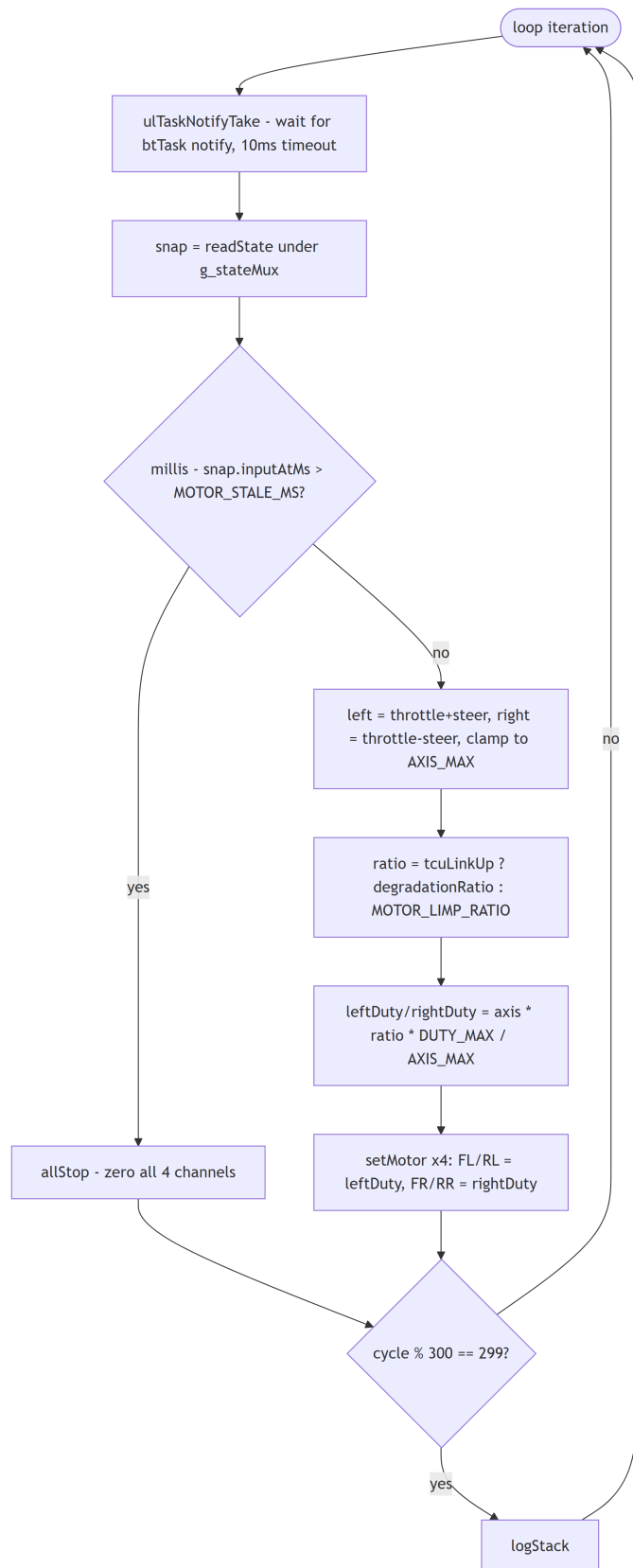
LinkState State Machine

SEARCH / PAIR / CONNECTED, recomputed every btTask iteration from anyControllerConnected(), pairingMode, and hasBondedController rather than an explicit transition table — this diagram shows the effective behavior.



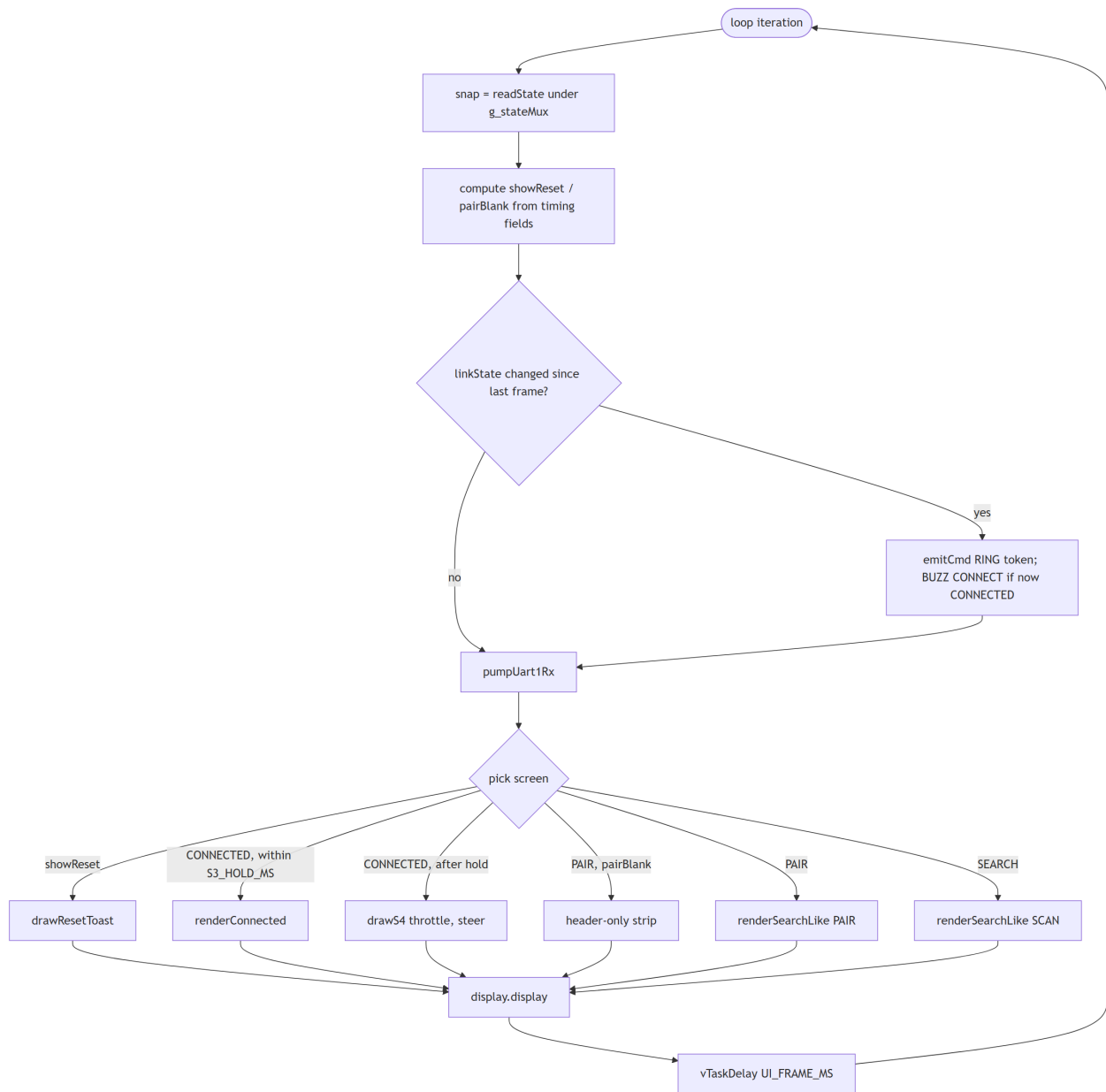
motorControlTask Loop (core 1, priority 5)

Wakes on btTask's notification (10 ms timeout fallback), takes a locked RobotState snapshot, zeros all motors if input is stale ($> \text{MOTOR_STALE_MS}$), otherwise mixes throttle/steer into left/right skid-steer duty, scales by the TCU degradation/limp ratio, and writes 20 kHz / 8-bit PWM to all four channels.



uiTask Loop (core 1, priority 1, ~6 Hz)

Takes a locked snapshot, emits RING/BUZZ on link-state change, pumps UART1 RX, then picks and draws one of the OLED screens (S1–S5) based on link state and timing flags.



End-to-End: Gamepad Stick → Motor PWM

The full data path from the physical stick axes, through the cross-core RobotState snapshot, to the skid-steer mix and PWM output driving the four motors.

